

Bab 2 - Integral Parsial

$$\frac{d}{dx} [u(x) \cdot v(x)] = \frac{d}{dx} [u(x)] \cdot v(x) + u(x) \cdot \frac{d}{dx} [v(x)]$$

$$u(x) \cdot \frac{d}{dx} [v(x)] = \frac{d}{dx} [u(x) \cdot v(x)] - \frac{d}{dx} [u(x)] \cdot v(x)$$

$$\int u(x) \frac{d}{dx} [v(x)] dx = \int \frac{d}{dx} [u(x) \cdot v(x)] dx - \int \frac{d}{dx} [u(x)] \cdot v(x) dx$$

$$\int u(x) \cdot dv(x) = u(x) \cdot v(x) - \int v(x) du(x)$$

$$\int u \, dv = uv - \int v \, du$$

misal

$$\int x \cos x \, dx$$

$\downarrow$ $\downarrow$
 u dv

misal:

$$u = x \quad dv = \cos x \, dx$$

$$\frac{du}{dx} = 1 \quad v = \sin x + C$$

$$du = dx$$

$$\int x \cos x \, dx = x \sin x - \int \sin x \, dx$$

$$= x \sin x - (-\cos x) + C$$

$$= x \sin x + \cos x + C$$

Contoh-2

$$\int x \ln x \, dx \Rightarrow = \frac{x^2 \ln x}{2} - \int \frac{x^2}{2} \cdot \frac{1}{x} dx$$

$$u = \ln x \quad dv = x \, dx \quad = \frac{x^2 \ln x}{2} - \frac{1}{2} \int x \, dx$$

$$\frac{du}{dx} = \frac{1}{x} \quad v = \frac{x^2}{2} + C \quad = \frac{x^2 \ln x}{2} - \frac{1}{2} \cdot \frac{1}{2} x^2 + C$$

$$du = \frac{1}{x} dx \quad v = \frac{x^2}{2} + C \quad = \frac{x^2 \ln x}{2} - \frac{1}{4} x^2 + C$$

Contoh-3:

$$\int \ln x \, dx \Rightarrow = x \ln x - \int x \frac{1}{x} dx$$

misal

$$u = \ln x \quad dv = dx \quad = x \ln x - \int dx$$

$$\frac{du}{dx} = \frac{1}{x} \quad v = \int dx = x \ln x - x + C$$

$$du = \frac{1}{x} dx \quad v = x + C$$

Contoh-4:

$$\int x^2 e^x \, dx$$

misal

$$u = x^2 \quad dv = e^x \, dx$$

$$du = 2x \, dx \quad v = e^x$$

$$\int x^2 e^x \, dx = x^2 e^x - \int e^x 2x \, dx$$

$$u = x^2 \quad dv = e^x \, dx$$

$$du = 2x \, dx \quad v = e^x + C$$

$$\int x^2 e^x \, dx = x^2 e^x - 2 \int e^x x \, dx$$

$$u = x \quad dv = e^x \, dx$$

$$\frac{du}{dx} = 1 \quad v = e^x + C$$

$$du = dx \quad v = e^x + C$$

$$\int x^2 e^x \, dx = x^2 e^x - 2 \left[x e^x - \int e^x \, dx \right]$$

$$= x^2 e^x - 2x e^x + 2e^x + C$$

Contoh-5:

$$\int e^x \sin x \, dx$$

$$u = e^x \quad dv = \sin x \, dx$$

$$\frac{du}{dx} = e^x \quad v = -\cos x + C$$

$$du = e^x \, dx$$

$$\int e^x \sin x \, dx = -e^x \cos x - \int -\cos x e^x \, dx$$

$$= -e^x \cos x + \int \cos x e^x \, dx$$

$$\int e^x \sin x \, dx = -e^x \cos x + e^x \sin x - \int e^x \sin x \, dx$$

$$2 \int e^x \sin x \, dx = -e^x \cos x + e^x \sin x$$

$$\int e^x \sin x \, dx = \frac{-e^x \cos x + e^x \sin x}{2} + C$$

2.2. Integrasi Fungsi Rasional

$$\frac{1}{x+1} + \frac{1}{x-2} = \frac{(x-2) + (x+1)}{(x+1)(x-2)} = \frac{2x-1}{x^2-x-2}$$

$$\begin{aligned} \int \frac{2x-1}{x^2-x-2} dx &= \int \frac{1}{x+1} + \frac{1}{x-2} dx \\ &= \int \frac{1}{x+1} dx + \int \frac{1}{x-2} dx \\ &= \ln|x+1| + \ln|x-2| + C \end{aligned}$$

Maka kita perlu cara
mengkomposisi Pecahan

1. Coba faktorkan penyebut

$$\frac{2x-1}{x^2-x-2} = \frac{2x-1}{(x+1)(x-2)}$$

2. dekomposisi jadi 2

$$\frac{2x-1}{x^2-x-2} = \frac{A}{x+1} + \frac{B}{x-2} \quad A, B \Rightarrow \text{konstanta}$$

$$\begin{aligned} \frac{2x-1}{x^2-x-2} &= \frac{A(x-2) + B(x+1)}{(x+1)(x-2)} \\ &= \frac{(A+B)x - 2A + B}{(x+1)(x-2)} \end{aligned}$$

$$2x-1 = (A+B)x - 2A+B$$

$$\begin{aligned} A+B &= 2 & A+B &= 2 \\ -2A+B &= -1 & B &= 1 \end{aligned}$$

$$\begin{aligned} 3A &= 3 \\ A &= 1 \end{aligned}$$

$$\frac{2x-1}{x^2-x-2} = \frac{1}{x+1} + \frac{1}{x-2}$$

Alternatif

$$2x-1 = A(x-2) + B(x+1)$$

$$\begin{aligned} x=2 & \quad x=-1 \\ 2(2)-1 = B(2+1) & \quad 2(-1)-1 = A(-1-2) + B(0) \\ 3 = 3B & \quad -3 = -3A \\ B=1 & \quad A=1 \end{aligned}$$

Contoh-1:

$$\int \frac{1}{x^2+x-2} dx$$

$$\begin{aligned} \frac{1}{x^2+x-2} &= \frac{1}{(x+2)(x-1)} \\ &= \frac{A}{x+2} + \frac{B}{x-1} \\ &= \frac{A(x-1) + B(x+2)}{(x+2)(x-1)} \end{aligned}$$

$$1 = A(x-1) + B(x+2)$$

$$\begin{aligned} x=1 & \quad x=-2 \\ 1 = A(1-1) + B(1+2) & \quad 1 = A(-2-1) + B(-2+2) \\ 3B = 1 & \quad 1 = -3A + 0 \\ B = 1/3 & \quad A = -1/3 \end{aligned}$$

$$\begin{aligned} \int \frac{1}{x^2+x-2} dx &= \int \frac{-1/3}{x+2} + \frac{1/3}{x-1} dx \\ &= -\frac{1}{3} \int \frac{1}{x+2} dx + \frac{1}{3} \int \frac{1}{x-1} dx \\ &= -\frac{1}{3} \ln|x+2| + \frac{1}{3} \ln|x-1| + C \end{aligned}$$

Aturan fungsi linier

Jika bentuk faktor $(ax+b)^n$ maka
harus diuraikan menjadi

$$\frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \dots + \frac{A_m}{(ax+b)^m}$$

Contoh:

$$\int \frac{2x+4}{x^3-2x^2} dx$$

$$\begin{aligned} \frac{2x+4}{x^3-2x^2} &= \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-2} \\ &= \frac{A(x^2)(x-2) + B(x)(x-2) + C(x^2)(x-2)}{(x^3-2x^2)} \\ &= \frac{A(x^3-2x^2) + B(x^2-2x) + C(x^3-2x^2)}{x^2(x-2)} \end{aligned}$$

$$2x+4 = A(x-2) + B(x-2) + Cx^2$$

$$x=0$$

$$2(0)+4 = B(0-2)$$

$$4 = -2B \rightarrow B = -2$$

$$x=2$$

$$2(2)+4 = C(2)^2$$

$$8 = 4C \rightarrow C = 2$$

$$x=1$$

$$2(1)+4 = A(1)(1-2) + B(1-2) + C(1)^2$$

$$6 = -A - 2(-1) + 2$$

$$6 = -A + 4$$

$$2 = -A$$

$$A = -2$$

$$\int \frac{2x+4}{x^3-2x^2} dx = \int \frac{-2}{x} + \frac{-2}{x^2} + \frac{2}{x-2} dx$$

$$= -2 \ln|x| - \frac{2}{x} + 2 \ln|x-2| + C$$

Absorpsi Faktor Kuadratik

Jika menemui bentuk $(ax^2+bx+c)^n$ yang tidak dapat difaktorkan, maka perlu di ubar.

$$\frac{A_1x+B_1}{(ax^2+bx+c)} + \frac{A_2x+B_2}{(ax^2+bx+c)^2} + \dots + \frac{A_mx+B_m}{(ax^2+bx+c)^m}$$

Contoh:

$$\frac{2x+1}{x^3+x} = \frac{2x+1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1}$$

$$\int \frac{2x+1}{x^3+x} dx$$

$$\frac{2x+1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1}$$

$$= \frac{A(x^2+1) + x(Bx+C)}{x(x^2+1)}$$

$$= \frac{A(x^2+1) + Bx^2 + Cx}{x(x^2+1)}$$

$$2x+1 = A(x^2+1) + Bx^2 + Cx$$

$$x=0$$

$$2(0)+1 = A(1)$$

$$1 = A$$

$$x=1$$

$$2(1)+1 = A(1^2+1) + B(1^2) + C(1)$$

$$3 = 2 + B + C$$

$$B+C = 1$$

$$x=-1$$

$$2(-1)+1 = A(1+1) + B(1) + (-1)C$$

$$-1 = 2 + B - C$$

$$-3 = B - C$$

$$B+C = 1$$

$$B+C = 1$$

$$B-C = -3$$

$$-1+C = 1$$

$$2B = -2$$

$$C = 2$$

$$B = -1$$

$$\int \frac{2x+1}{x^3+x} dx = \int \frac{1}{x} + \frac{-x+2}{x^2+1} dx$$

$$= \int \frac{1}{x} dx + \int \frac{-x}{x^2+1} dx + \int \frac{2}{x^2+1} dx$$

$$= \ln|x| - \frac{1}{2} \ln|x^2+1| + 2 \tan^{-1}(x) + C$$

Integral Fungs: Rasional Tak Sejat

Jika menemui integral fungsi dengan derajat pembilang >= derajat penyebut. Caranya gunakan pembagian bersusun

Contoh:

$$\frac{x^4+x^2+1}{x^2+2x-3} = \dots = x^2 - 2x + 8 + \frac{-22x+25}{x^2+2x-3}$$

$$\begin{array}{r} x^2-2x+8 \\ x^2+2x-3 \overline{) x^4+x^2+1} \\ \underline{x^4+2x^3-3x^2} \\ -2x^3+4x^2+1 \\ \underline{-2x^3-4x^2-6x} \\ 8x^2-6x+1 \\ \underline{8x^2+16x-24} \\ -22x+25 \end{array}$$

$$\int \frac{x^3}{x^2-3x+2} dx$$

$$\begin{array}{r} x+3 \\ x^2-3x+2 \overline{) x^3} \\ \underline{x^3 - 3x^2 + 2x} \\ 3x^2 - 2x \\ \underline{3x^2 - 9x + 6} \\ 7x - 6 \end{array}$$

$$\int \frac{x^3}{x^2-3x+2} dx = \int x+3 dx + \int \frac{7x-6}{x^2-3x+2}$$

$$\frac{7x-6}{x^2-3x+2} = \frac{A}{x-1} + \frac{B}{x-2}$$

$$= \frac{A(x-2) + B(x-1)}{(x-1)(x-2)}$$

$$7x-6 = A(x-2) + B(x-1)$$

$$x=2 \rightarrow 7(2)-6 = B(2-1)$$

$$B=8$$

$$x=1 \rightarrow 7(1)-6 = A(1-2)$$

$$A=-1$$

$$= \int x+3 dx + \int \frac{-1}{x-1} dx + \int \frac{8}{x-2} dx$$

$$= \frac{1}{2}x^2 + 3x - \ln|x-1| + 8\ln|x-2| + C$$